

Introduction

When the sample size for a study is constrained by budget, time, or availability of participants, the natural planning question is no longer “what n achieves target power?” but instead “given my fixed n , what is the smallest effect I can reasonably hope to detect?” The minimum detectable effect (MDE) reverses the usual sample-size calculation, solving for the effect size at which power equals the target — typically 80 % — given the fixed n , α , and analysis. Reporting an MDE is increasingly required for grant applications with constrained sample sizes, for pilot and exploratory studies, and for honest power-analysis reporting where a study cannot afford to enrol enough participants for the original target.

Prerequisites

A working understanding of statistical power, the relationship between α , effect size, sample size, and power, and the standard test-specific effect-size measures (Cohen’s d , r , f^2 , w).

Theory

Power, effect size, significance level, and sample size are linked by the non-central distribution of the test statistic, and solving for any one of these quantities given the other three is the standard inversion in any power-analysis framework. The MDE is defined as the effect size at which the planned test achieves a specified power (usually 0.80) under the planned α and the available sample size. Plotted as a curve of MDE vs. n , this gives a clear picture of the design’s sensitivity across resource scenarios.

Assumptions

The same assumptions as the underlying test apply (Normality for t -tests, bivariate Normality for correlation, large expected counts for chi-squared, etc.); the MDE is a property of the planned analysis under the design, not a property of the observed data after the fact.

R Implementation

```
library(pwr)

pwr.t.test(n = 40, sig.level = 0.05, power = 0.80,
           type = "two.sample", d = NULL)

pwr.r.test(n = 100, sig.level = 0.05, power = 0.80)

n_grid <- seq(10, 200, by = 5)
d_grid <- sapply(n_grid, function(n)
  pwr.t.test(n = n, sig.level = 0.05, power = 0.80, type = "two.sample")$d)

plot(n_grid, d_grid, type = "l", lwd = 2, col = "#2A9D8F",
     xlab = "n per group", ylab = "MDE (Cohen's d)",
     main = "Minimum detectable effect at power = 0.80")
```

Output & Results

For a two-sample t -test with $n = 40$ per group, the MDE at 80 % power is $d = 0.64$ (medium-to-large); doubling to $n = 200$ per group reduces the MDE to $d = 0.28$ (small-to-medium). For a correlation test with $n = 100$, the MDE is $r = 0.28$. Plotting the MDE-vs- n curve shows the diminishing-returns relationship that planners should understand when negotiating sample-size constraints.

Interpretation

A reporting sentence: “With the available sample of 40 participants per arm, the minimum detectable standardised effect (Cohen’s d) at 80 % power and two-sided $\alpha = 0.05$ is 0.64 — a medium-to-large effect by Cohen’s conventions. Effects smaller than this magnitude would likely remain undetected. The smallest clinically meaningful effect for the primary outcome is $d = 0.40$, so the planned study is underpowered for the clinical target; we report the MDE explicitly to clarify the inferential limits and propose a larger follow-up trial.” Always discuss MDE relative to the clinically meaningful effect.

Practical Tips

- Report the minimum detectable effect explicitly when sample size is constrained; this prevents inflated claims of null effects (“no significant difference”) when the study was simply too small to detect a clinically meaningful effect, a recurring problem in the literature.
- Couple MDE reporting with explicit discussion of whether the MDE is clinically or scientifically meaningful; an MDE that exceeds the smallest meaningful effect tells readers the study could not have detected the truth even if it were present.
- For exploratory pilot studies, report MDE in place of formal hypothesis testing; pilot studies are not powered for inference and the MDE conveys the design’s sensitivity honestly.
- MDE inflates rapidly for sub-group analyses (smaller n per cell), cluster-adjusted analyses (larger effective standard errors), repeated-measures designs with low correlation, and any setting where the effective sample size is reduced from the nominal n . Account for these factors in the MDE calculation, not just the headline figure.
- Simulation-based MDE estimation is recommended for complex designs (mixed-effects models, longitudinal data with informative dropout, generalised linear models with heavy-tailed outcomes) where the standard non-central-distribution formulas do not apply directly.
- Pre-register the MDE in the protocol or analysis plan; reporting MDE only after the fact when results turn out non-significant is methodologically suspect, while a pre-registered MDE protects against that interpretation.

R Packages Used

`pwr` family of functions with the effect-size argument set to `NULL` to solve for MDE; `pwrss` for extended MDE calculations across many test types; `WebPower` for an alternative interface; `simr::powerSim()` for simulation-based MDE in mixed-effects designs; `Superpower` for ANOVA-based MDE simulations.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation